

Nonlinear optical properties of calcium barium niobate epitaxial thin films

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Abstract: We investigate the potential of epitaxial calcium barium niobate (CBN) thin film grown by pulsed laser deposition for optical frequency conversion. Using second harmonic generation (SHG), we analyze the polarization response of the generated signal to determine the ratios d_{15} / d_{32} and d_{33} / d_{32} of the three independent components of the second-order nonlinear susceptibility tensor in CBN thin film. In addition, a detailed comparison to the signal intensity obtained in a y-cut quartz allows us to measure the absolute value of these components in CBN thin film: $d_{15} = 5 \pm 2$ pm / V, $d_{32} = 3.1 \pm 0.6$ pm / V and $d_{33} = 9 \pm 2$ pm / V.

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OCIS codes: (190.2620) Harmonic generation and mixing; (190.4400) Nonlinear optics, materials; (190.4160) Multiphoton process; (160.4330) Nonlinear optical materials; (310.6860) Thin films, optical properties.

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1. Introduction

Ferroelectrics materials exhibit unique properties. In particular, they show high performance for optics and are commonly used for optical modulation [1], light generation [2,3] optical filtering [4] and many more applications. These properties come from their strong intrinsic polarization that can be tuned by applying an external excitation such as an electric field, pressure or even a strong optical pulse [5–7]. However, if some of these ferroelectrics are used in their bulk form like Lithium Niobate (LiNbO_3 , LN) [8], very few have been successfully grown as a thin film retaining their capacity for frequency conversion, optical modulation or dielectric tunability.

These few include materials from the Tetragonal Tungsten Bronze family such as Strontium Barium Niobate ($\text{Sr}_x\text{Ba}_{1-x}\text{Nb}_2\text{O}_6$, SBN). In particular, this material has shown unique nonlinear optical properties such as second [9] and third harmonic generation [10] or even Cerenkov nonlinear interactions [11]. However, it suffers from a low Curie temperature that limits its use in practical devices. As the material is likely to be exposed to high optical power in integrated devices, it will undergo a phase transition that makes its nonlinear response changes [12]. In this context, a similar material but with higher Curie temperature (about 200°C higher) was recently developed and studied, namely Calcium Barium Niobate ($\text{Ca}_x\text{Ba}_{1-x}\text{Nb}_2\text{O}_6$, CBN [13]) that has already shown nonlinear properties in its bulk form such as Cerenkov-type Second Harmonic Generation [14] as well as broadband optical frequency conversion [15]. This promising new material was grown as a crystalline thin film using different deposition techniques [16] with a calcium ratio of 28%mol corresponding to the most stable composition ($x = 0.28$, $\text{Ca}_{0.28}\text{Ba}_{0.72}\text{Nb}_2\text{O}_6$). It shows a very strong potential as an integrated nonlinear prism or optical frequency converter. However, in order to fully exploit its optical properties, it is necessary to characterize precisely the origin of this nonlinearity such as the d_{ij} components of its second-order nonlinear susceptibility tensor.

In this paper, we investigate the nonlinear optical properties of epitaxial CBN thin films grown by Pulsed Laser Deposition (PLD) on MgO substrate using the operating conditions

reported in [16]. The epitaxy was confirmed by X-Ray Diffraction. In particular, we measured for the first time all the components of the second order nonlinear susceptibility in CBN thin films. This study will constitute the basis of the fabrication of nonlinear optical integrated devices based on this material.

2. Material and methods

2.1 CBN thin films

The CBN thin films were deposited on MgO (001) double-side polished substrates using a CBN-28 commercial ceramic target from K.J. Lesker. The deposition is carried out in the PVD 3000 deposition system of PVD Products at 800°C with an oxygen background pressure of 1 mTorr. The KrF (248 nm) laser is focused on the target with a fluence of 2 J/cm². After deposition, the deposited CBN thin film is annealed *in situ* at 800°C for 30 minutes in 300 mTorr of oxygen to reduce the oxygen vacancies. For a more detailed description of the deposition process see [16].

2.2 SHG and I-SHG microscopy

The CBN thin film was first imaged using a laser scanning multiphoton microscope. For a complete description see [17]. Briefly, the laser source is a Titanium:Sapphire oscillator (Tsunami, Spectra Physics) delivering 150 fs pulses at a 80 MHz repetition rate with the central wavelength tuned to 810 nm. The laser was sent on two galvanometric mirrors and tightly focused on the sample using a high numerical aperture objective (20x, 0.75NA, Olympus). Images were acquired by scanning the focal point on the sample. SHG signals were collected through a condenser and the wavelength of interest (405 nm) was selected using two low-pass filters (FF01-720/SP-25, Semrock) and a narrow bandpass filter (FF01-405/10, Semrock). Finally, the SHG signals were detected by a photomultiplier tube (R6357, Hamamatsu) biased at about 350 V. In these experiments, the average exciting power was set to about 40 mW, which correspond to 0.5 nJ per pulse.

In addition, to investigate the orientation of the ferroelectric domains in the CBN thin film we used Interferometric Second Harmonic Generation microscopy (I-SHG). For a complete description of the setup see [18]. Both SHG and I-SHG techniques were implemented on the same setup in order to be able to switch rapidly between these two modes. In short, I-SHG is a modification of a standard SHG microscope that allows to retrieve the phase of the signal generated in the sample by measuring the interferences with a reference SHG beam, generated in a y-cut quartz plate, outside the microscope. A 1.5 mm thick BK7 glass window, placed on a rotating mount, was added to control the phase between the reference and the sample SHG beams on the detector. To isolate the interferometric term, we computed the difference between two images acquired with a π phase-shift in the reference phase. Finally, the acquisition of 18 pairs of images with references phases ranging from 0° to 330° by 30° step, allows us to extract the sample SHG phase in every pixel.

2.3 Characterization of the SHG polarization response

To further investigate the nonlinear properties of CBN thin film, another setup was implemented, allowing to illuminate the sample in the different configurations required to probe the independent components of the nonlinear susceptibility tensor. In these experiments, the Ti:Sapph laser (150 fs pulses length, 810 nm wavelength, 80 MHz repetition rate, approximately 200 mW average power corresponding to about 2.5 nJ per pulse) is directly focused on the CBN thin film (750 nm thick) using a 5 cm focal lens. The incident beam is linearly polarized. A second lens placed after the sample is used to collimate both fundamental and SHG beams. The pump beam is then filtered using two low-pass filters and a narrow bandpass filter (FF01-720/SP-25 and FF01-405/10, Semrock), and the SHG intensity is detected on a CCD camera. The incident light polarization is rotated using a half-waveplate

and the polarization reaching the detector is selected using a polarizer set before the camera. Finally, the CBN thin film is placed on a rotating mount to probe the out-of-plane components of the tensor.

3. Results and discussion

3.1 Structure of the CBN thin films

We first study the structure of the deposited CBN thin films using X-Ray Diffraction (XRD). As can be seen on Fig. 1(a), only the (001) family of planes can be seen for the CBN thin film in addition to the MgO substrate main peak. It confirms that the out-of-plane orientation of CBN is its c-axis. As for the in-plane orientation, Fig. 1(b) shows the typical CBN phi-scan for the (311) plane. Two families of orientations are observed for CBN, tilted of $+30.5^\circ$ and -30.5° with respect to the substrate. More details can be found in [16,19]. Both XRD and Phi-scan confirm the epitaxial nature of the deposited CBN thin film on the MgO substrate.

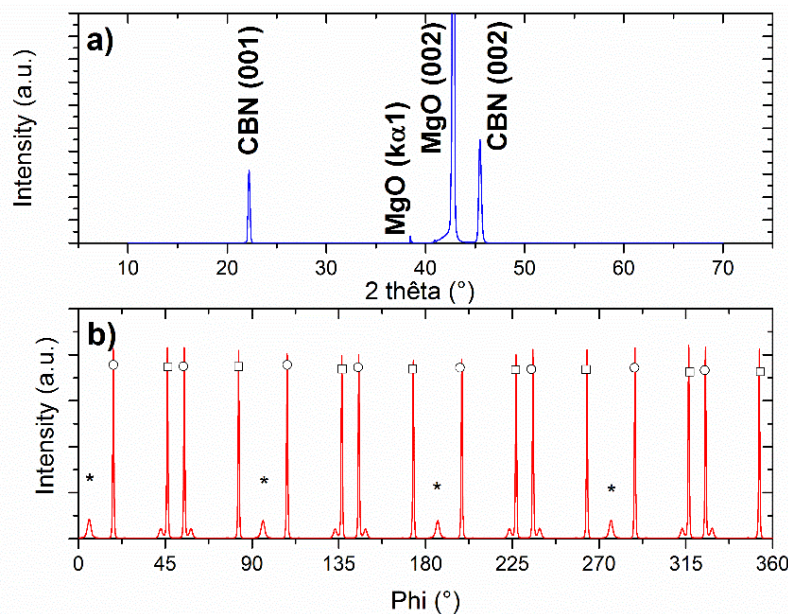


Fig. 1. (a) XRD theta-2theta scan of the deposited CBN thin film on MgO using pulsed laser deposition at 2 J/cm^2 , 1 mTorr O_2 pressure and 800°C substrate temperature. The thin film was annealed at 800°C under 300 mTorr O_2 pressure for 30 min *in situ* after deposition. (b) Phi-scan of the oblique (311) plane of the CBN thin film. The stars represent the (110) orientation of the MgO substrate. The circles are for the $+30.5^\circ$ and the squares for the -30.5° orientations of the (311) plane of the CBN thin film.

3.2 SHG in CBN thin films

Figure 2(a) shows the SHG image of the CBN thin film sample. The intensity variations are attributed to the roughness of the unpolished sample surface (7nm rms [19]). Figure 2(b) displays the square root of the second harmonic intensity detected by the photomultiplier tube, both in trans- and epi-directions, as a function of the power of the incident laser beam. The linear behavior emphasizes the second order origin of this process. Notably, since the coherence length in the backward direction is small compared to the sample thickness, the signal measured in the epi-direction corresponds to backreflection of the forward emitted signal.

Using I-SHG microscopy, we measured the phase of the SHG signal generated in the CBN thin film, which allows to probe the polarity of the sample. Indeed, even if the XRD measurements demonstrate the epitaxial nature of the thin film, the sample might be

polydomains, meaning that the orientation of the spontaneous polarization could change randomly between $\pm \pi/2$. This might have an impact on the phase matching [14,15].

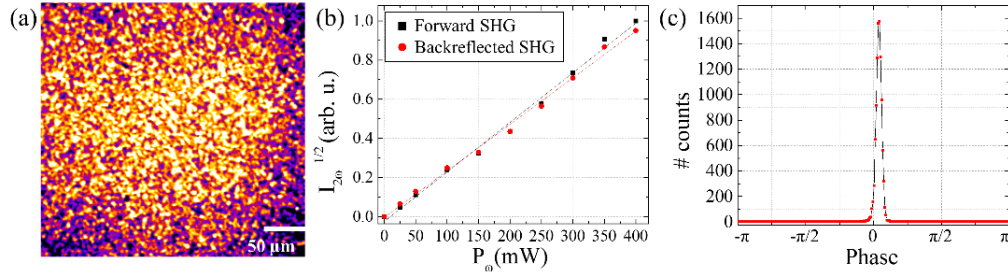


Fig. 2. (a) SHG image of the CBN thin film. (b) Square root of the SHG intensity as a function of the excitation power. (c) Phase histogram measured in a $100 \times 25 \mu\text{m}^2$ image of the sample.

Figure 2(c) shows the phase histogram from a $100 \times 25 \mu\text{m}^2$ region of the sample. The very peaked distribution indicates that only one polarity is present in the ferroelectric phase, showing that our sample is mono-domain, at least within an imaging plane of $2500 \mu\text{m}^2$.

3.3 Ratios of the independent nonlinear susceptibility components

The C_{4v} symmetry of the ferroelectric phase of CBN imposes the form of the nonlinear susceptibility tensor:

$$d = \begin{pmatrix} 0 & 0 & 0 & 0 & d_{15} & 0 \\ 0 & 0 & 0 & d_{24} & 0 & 0 \\ d_{31} & d_{32} & d_{33} & 0 & 0 & 0 \end{pmatrix}. \quad (1)$$

with $d_{31} = d_{32}$ and $d_{15} = d_{24}$.

In the case of a low numerical aperture lens, as used here, the excitation beam can be approximated by plane waves propagating along z axis, with their electric field orthogonal to this direction. This incident wave produces a second order nonlinear polarization $P^{(2)}$:

$$\begin{pmatrix} P_x^{(2)} \\ P_y^{(2)} \\ P_z^{(2)} \end{pmatrix} \propto \begin{pmatrix} -d_{15}c_\varphi^2s_\theta c_\theta^2 - d_{32}c_\varphi^2c_\theta^2s_\theta - d_{32}s_\varphi^2s_\theta - d_{33}c_\varphi^2s_\theta^3 \\ -d_{15}s_\varphi c_\varphi s_\theta \\ -d_{15}c_\varphi^2c_\theta s_\theta^2 + d_{32}c_\varphi^2c_\theta^3 + d_{32}s_\varphi^2c_\theta + d_{33}c_\varphi^2s_\theta^2c_\theta \end{pmatrix}. \quad (2)$$

where φ is the angle between the laser polarization and the x -axis and θ the angle between the sample and the focal plane $\{xy\}$ (see Fig. 3). Here, c_i and s_i stand for the cosine and sine of the angle i respectively. Therefore, only two configurations of incident and detected polarizations are required to extract the different tensor components.

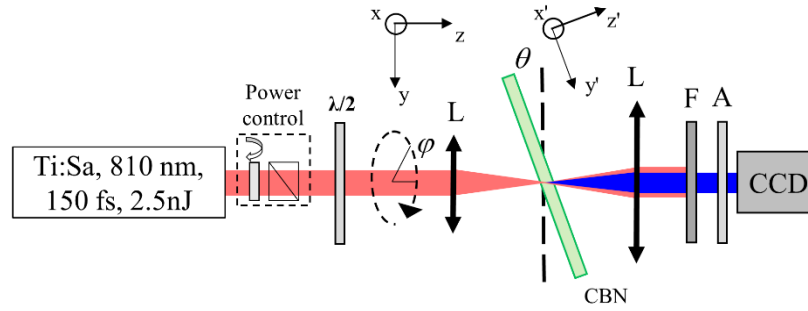


Fig. 3. Experimental setup. $\lambda/2$: half-waveplate, L: lens, F: filters and A: analyzer.

First, for a given direction of the incident polarization ($\varphi = 0^\circ$), the CBN thin film was rotated around θ from 0° to 60° , and the SHG signal was detected with the analyzer along the x axis.

$$I_x(\varphi_0, \theta) \propto \left[c_\theta^2 s_\theta + \frac{d_{15}}{d_{32}} c_\theta^2 s_\theta + \frac{d_{33}}{d_{32}} s_\theta^3 \right]^2. \quad (3)$$

Second, for a given thin film angle ($\theta_0 = 20^\circ$), the incident polarization (φ) was rotated while the SHG signal was detected with the analyzer along the x axis.

$$I_x(\varphi, \theta_0) \propto \left[s_\varphi^2 s_{\theta_0} + c_\varphi^2 c_{\theta_0}^2 s_{\theta_0} + \frac{d_{15}}{d_{32}} c_\varphi^2 c_{\theta_0}^2 s_{\theta_0} + \frac{d_{33}}{d_{32}} c_\varphi^2 s_{\theta_0}^3 \right]^2. \quad (4)$$

Figure 4 displays the results of the measurements in these two configurations, using the setup shown in Fig. 3. The full squares represent the intensity measurements, corrected for the variation of the transmission and reflection coefficients, both at the CBN and substrate interfaces, while changing the incident angle θ and the incident polarization direction φ .

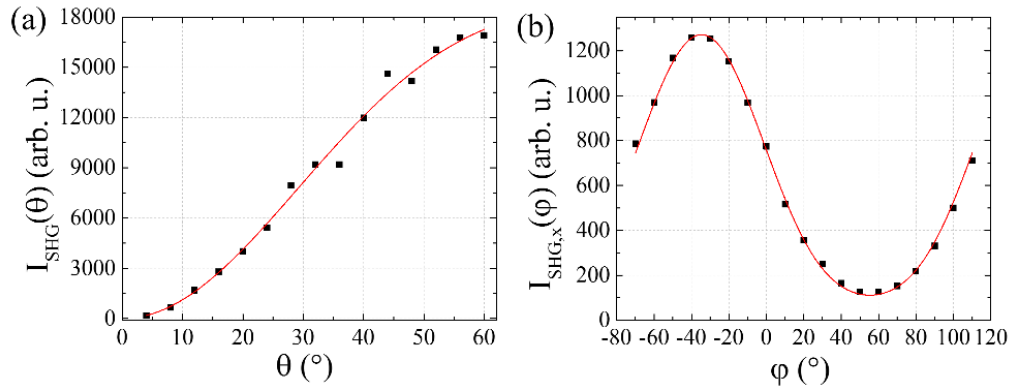


Fig. 4. (a) SHG intensity as a function of the angle between the CBN thin film and the focal plane $\{xy\}$ for a fixed direction of the incident laser polarization φ_0 . (b) SHG intensity as a function of the polarization angle for a fixed polar angle θ_0 of the sample. Black squares represent experimental measurement while red straight lines show fits using Eq. (3) and (4) respectively.

The red lines in Fig. 4(a) and 4(b) represents the fitting of the experimental data using Eqs. (3) and (4) respectively. These fits were used to extract the two ratios of the independent tensor components $d_{15}/d_{32} = 1.6 \pm 0.3$ and $d_{33}/d_{32} = 3.0 \pm 0.2$. The out-of-plane component d_{33} is the highest as expected from CBN growth orientation [19] and crystal class (C_{4v}) [12], and was also noticed previously in bulk CBN [15]. Interestingly, it is found that d_{15} is quite higher

than d_{32} , which indicates that the Kleinman symmetry, allowing to permute the indices, does not apply here [20]. Indeed, since the band gap of CBN is about 3.45 eV [16], corresponding to a wavelength of 359 nm, the second harmonic frequency (405 nm) is close to a resonance which causes this significant deviation from the Kleinman condition at these wavelengths.

3.4 Absolute determination of the nonlinear susceptibility

Finally, the absolute value of the nonlinear component was measured by comparing the SHG power obtained from CBN thin film with that obtained from a 350 μm y-cut quartz plate with the same excitation parameters. Again, since a low numerical aperture lens is used here, the confocal parameter is about 1 mm, so that we can approximate the phase-matching factor by the usual plane wave approximation. Therefore, the SHG power $P_{2\omega}$ is given by [20]:

$$P_{2\omega} = \frac{2\sqrt{2}\pi P_{\omega}^2}{n_{2\omega} n_{\omega}^2 c \epsilon_0 \lambda^2 R \tau w^2} d_{\text{eff}}^2 \left(L \sin c \left(\frac{L}{L_c} \right) \right)^2. \quad (5)$$

where $n_{2\omega}$ (resp. n_{ω}) is the refractive indices at 405 nm (resp. 810 nm), P_{ω} the average incident power, c the speed of light, ϵ_0 the vacuum permittivity, λ the laser wavelength, R the repetition rate, τ the pulse duration, w the surface of focalization, d_{eff} the effective nonlinear susceptibility, depending on the geometry of excitation, L the sample thickness and L_c the coherence length (Table 1).

Table 1. Ordinary refractive index [16,21,22] and coherence length of CBN thin film, MgO substrate and reference quartz (c, m and q superscript respectively) along with transmission coefficient of the sample (at $\theta = 20^\circ$).

n_{ω}^c	$n_{2\omega}^c$	n_{ω}^m	$n_{2\omega}^m$	n_{ω}^q	$n_{2\omega}^q$	L_c^c (μm)	L_c^q (μm)	T^q	T^c
2.25	2.50	1.73	1.76	1.54	1.56	0.53	6.82	0.94	0.69

Using Eq. (5) for CBN and quartz, and correcting for the different transmission coefficients (T) in the thin film and the quartz plate, the effective nonlinear susceptibility can be deduced. Rotating the thin film from $\theta = 20^\circ$ and detecting the signal with the analyzer along the y-axis provides the relationship between d_{32} and the nonlinear susceptibility of quartz $d_{11} = 0.4 \text{ pm/V}$ [23,24], using the following equation:

$$\left(\frac{d_{32}^c \sin \theta}{d_{11}^q} \right)^2 = \frac{n_{2\omega}^q n_{\omega}^{q^2}}{n_{2\omega}^c n_{\omega}^{c^2}} \frac{T^q}{T^c} \frac{P_{2\omega}^c}{P_{2\omega}^q} \frac{L^{q^2} \sin^2 c \left(L^q / L_c^q \right)}{L^{c^2} \sin^2 c \left(L^c / L_c^c \right)}. \quad (6)$$

where the superscripts c and q stand for the CBN thin films and the quartz respectively, and l_{θ} is the effective thickness of the CBN thin film at the angle θ . It is worth noting that T^c takes into account the reflexion on both the MgO substrate and the CBN thin film.

Finally, using the ratios d_{15}/d_{32} and d_{33}/d_{32} previously measured, we can obtain each of the three independent components of CBN as reported in Table 2.

Table 2. Nonlinear susceptibility components of CBN thin film

	d_{15}	d_{32}	d_{33}
d (pm/V)	5.0 ± 2	3.1 ± 0.4	9 ± 2

These values are in good agreement with those previously estimated from comparison with SBN crystal [15,25]. In addition, the absolute values well compare with those of other common ferroelectric materials in their bulk form such as KTP ($d_{33} = 13.7 \text{ pm/V}$), KD*P ($d_{16} = 1.8 \text{ pm/V}$), BBO ($d_{16} = 1.8 \text{ pm/V}$) or Lithium Niobate ($d_{15} = -5.95 \text{ pm/V}$, $d_{33} = 31.5 \text{ pm/V}$) [26].

4. Conclusion

We have studied the efficiency of epitaxial ferroelectric CBN thin film for optical frequency conversion. Using polarization-resolved SHG, the second-order nonlinear susceptibility components of the sample were probed. The classical approximation of Kleinman symmetry was shown to be inapplicable here, which implies that the second-order nonlinear susceptibility tensor of CBN has three independent components. Finally, comparing our results to a reference quartz plate, the nonlinear susceptibility of CBN was measured for the first time. This study contributes to a better characterization of the very promising properties of CBN, which is a good candidate for integrated nonlinear prism or optical frequency converter for on-chip devices.

Acknowledgments

The author acknowledges the financial support from the Canada Foundation for Innovation (CFI), the Natural Sciences and Engineering Research Council of Canada (NSERC), and the Fond Québécois de la Recherche sur la Nature et les Technologies (FQRNT).